

Machine Learning - Graduate Level Assessment 6

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

Statement 1| ImageNet has images of various resolutions. Statement 2| Caltech-101 has more images than ImageNet.

- A. True, True
- B. False, False
- C. True, False
- D. False, True

Question 2

(2 marks)

Statement 1| Word 2 Vec parameters were not initialized using a Restricted Boltzman Machine. Statement 2| The tanh function is a nonlinear activation function.

- A. True, True
- B. False, False
- C. True, False
- D. False, True

Question 3

(2 marks)

What is the dimensionality of the null space of the following matrix? $A = \begin{bmatrix} 3 & 2 & -9 \\ -6 & -4 & 18 \\ 12 & 8 & -36 \end{bmatrix}$

- A. 0
- B. 1

C. 2

D. 3

Question 4

(2 marks)

After applying a regularization penalty in linear regression, you find that some of the coefficients of w are zeroed out. Which of the following penalties might have been used?

A. L 0 norm

B. L 1 norm

C. L 2 norm

D. either (a) or (b)

Question 5

(2 marks)

Averaging the output of multiple decision trees helps _.

A. Increase bias

B. Decrease bias

C. Increase variance

D. Decrease variance

True / False

Question 1

(2 marks)

Residual connections are commonly utilized in both ResNets and Transformers to facilitate training of deep networks.

True

False

Question 2*(2 marks)*

The joint probability $P(A, B, C)$ can be expressed as $P(A, B | C) * P(C)$.

Question 3*(2 marks)*

Underfitting occurs when a model fails to capture the underlying patterns in the training data and cannot generalize to new data.

Question 4*(2 marks)*

The method used to learn weights, such as matrix inversion or gradient descent, primarily affects the trade-off between underfitting and overfitting.

Question 5*(2 marks)*

Clustering techniques are effective for detecting fraudulent credit card transactions by grouping similar behaviors.

Long Form**Question 1***(10 marks)*

Discuss the validity of the statements regarding entropy in joint distributions and the effects of moralization on directed graphs. Analyze their implications in machine learning contexts.

Discuss the validity of the statements regarding entropy in conditional distributions and the effects of moralization on directed graphs. Analyze their implications in machine learning contexts.

Type your answer here...

Question 2

(10 marks)

Discuss the assumptions of the Naive Bayes classifier, specifically addressing the implications of attribute independence given the class value. Why is this assumption critical?

Type your answer here...

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